

# Jacinta Wayua Muindi

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## Summary

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Statistician and data scientist with an MSc in Statistics and a strong foundation in Bayesian/probability modelling, machine learning, time series analysis, and stochastic processes, among others. Author of two peer-reviewed publications spanning non-linear time-series analysis and causal inference methodology. Experienced in Python and R, with hands-on exposure to large language models, NLP pipelines, and ensemble ML methods gained through industry and research collaborations. Driven by a desire to translate these technical capabilities into meaningful impact, with a particular interest in applying data-driven approaches to real-world challenges in public health and beyond.

## Work Experience

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**Part-Time Statistics Lecturer**, University of Nairobi, Nairobi Mar 2024 – Present

- I develop and deliver lectures in statistics to undergraduate students, fostering a dynamic learning environment conducive to critical thinking and analytical skills development
- I develop course materials, assignments, and exams, and also invigilate exams
- I mentor and guide students, igniting interest in statistics and statistical tools through personalized support and hands-on experience.
- **Courses Taught:** Introduction to Probability & Statistics, Probability & Statistics II, Biostatistics.

**Compliance Officer/MLRO**, Pearl Forex Bureau, Nairobi Apr 2024 – Jan 2026

- I led data-driven transaction monitoring and analysis to detect unusual activity and assess financial crime risks, ensuring insights inform compliance decisions and reporting. I also oversaw the implementation of AML/CFT/CPF policies, development of sound internal controls to ensure compliance, and supported staff training and regulatory alignment.

**Machine Learning Collaborator**, OdumareTech Feb 2024 – May 2024

- Collaborated on a project to develop, evaluate, and deploy a Machine Learning system for hate speech detection by utilizing different ML algorithms, aiming to develop a tool that could alleviate the problem of hate speech on social media platforms.

**Data Science Intern**, ExploreAI Aug 2023 – Dec 2023

- Collaborated on a project to develop and implement an ML/AI model, where we enhanced content alignment with specific user personas for the marketing team, using a pretrained LLM (Llama-2-7b) and frameworks like Langchain and PyTorch.
- Contributed to projects on data analysis using Python and SQL, successfully extracting actionable insights from extensive datasets, and leveraged Power BI to create compelling visualizations and dashboards to effectively communicate key findings and enhance decision-making.

**Census Content Supervisor**, Kenya National Bureau of Statistics Aug 2019 – Sept 2019

- Coordinated and conducted successful census operations including enumerator training, logistical preparation, enumeration areas identification, supervision, problem solving, re-interviews, and data review, ensuring accurate and efficient data collection.

## Education

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**University of Nairobi**, Master of Science in Statistics Sept 2021 – Dec 2023

- **Research Project:** Self-Exciting Threshold Autoregressive Modelling of the NSE 20 Share Index Using the Bayesian Approach

**Technical University of Kenya**, Bachelor of Science in Mathematics Jan 2013 - Dec 2016

- Second Class Honours (Upper Division)

**Machakos Girls' High School**, Kenya Certificate of Secondary Education (KCSE)

Feb 2008 – Dec 2011

- Grade: B+

## Trainings & Certification

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### **Foundational Methods in Data Science - AIMS Research and Innovation Centre**

Mar 2026 - Apr 2026

Learned foundational approaches in data science drawn from: Statistics and optimisation, Linear algebra and numerical analysis, Probability theory and functional analysis, Machine learning, and artificial intelligence.

### **Certificate of Completion for Data Science Course - Explore AI Academy**

Jan 2023 - Dec 2023

Learned Analytic Programming, Data Visualization, Machine Learning, Cloud Computing, and Model Deployment using Streamlit and Flask.

### **Data Analytics Packages by Mathematics Association of Nairobi University**

May 2022 - Aug 2022

Learned Data Management using Advanced Excel, Statistical Analysis using R-Programming, and Data Analysis using Python and SPSS.

## Skills

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**Hard Skills:** Machine Learning Modeling, Statistical Analysis and Modeling, Data Analysis, Natural Language Processing (NLP)/LLMs, Data Visualization, Version Control

**Languages and Tools:** R, Python, PowerBI, SQL, SPSS, Excel, Tableau, GitHub

**Soft Skills:** Analytical Thinking, Problem-Solving, Good Verbal and Written Communication, Stakeholder Engagement, Collaboration, Adaptability, Attention to Detail, Time Management.

## Publications

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### **Self-Exciting Threshold Autoregressive (SETAR) Modelling of the NSE 20 Share Index Using the Bayesian Approach**

Nov 2024

American Journal of Theoretical and Applied Statistics (Volume 13, Issue 6)  
10.11648/j.ajtas.20241306.13

### **Propensity Score Matching: An Application on Observational Data**

Dec 2024

American Journal of Mathematical and Computer Modelling (Volume 9, Issue 3)  
10.11648/j.ajmcm.20240903.12

## Workshops

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### **East-African School for Young Researchers on Advanced Machine Learning Techniques (EASY-ML)**

- By Centre National De La Recherche Scientifique (CNRS) and University of Nairobi in July 2025

### **AFRICAST Workshop on Tidy Time Series & Forecasting in R**

- By the International Institute of Forecasters in October 2023

### **School on Mathematical Programming and Algorithms**

- By ICTP-EAUMP in July 2022

## Professional affiliations

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**Kenya Women in Mathematical Sciences Association**